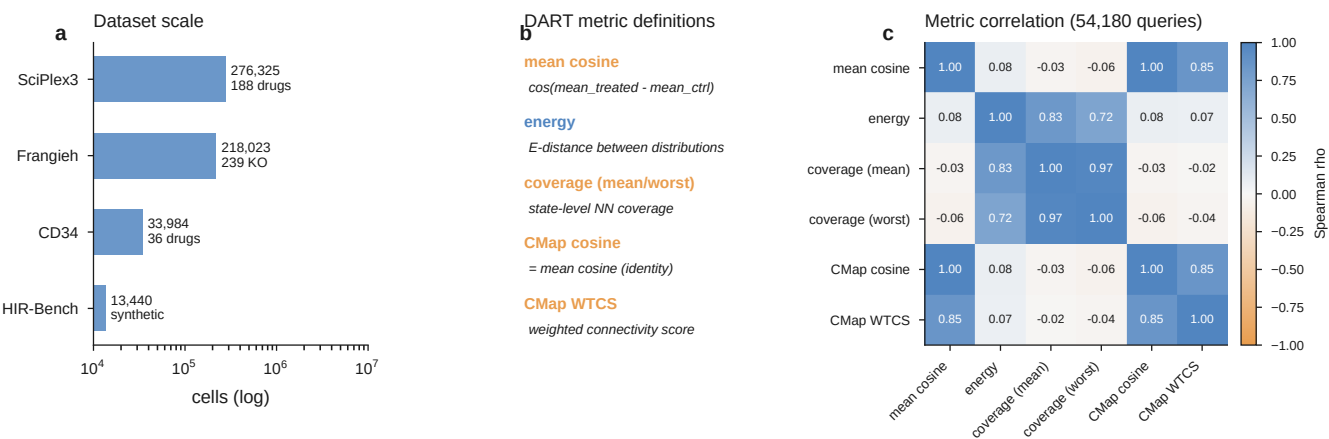


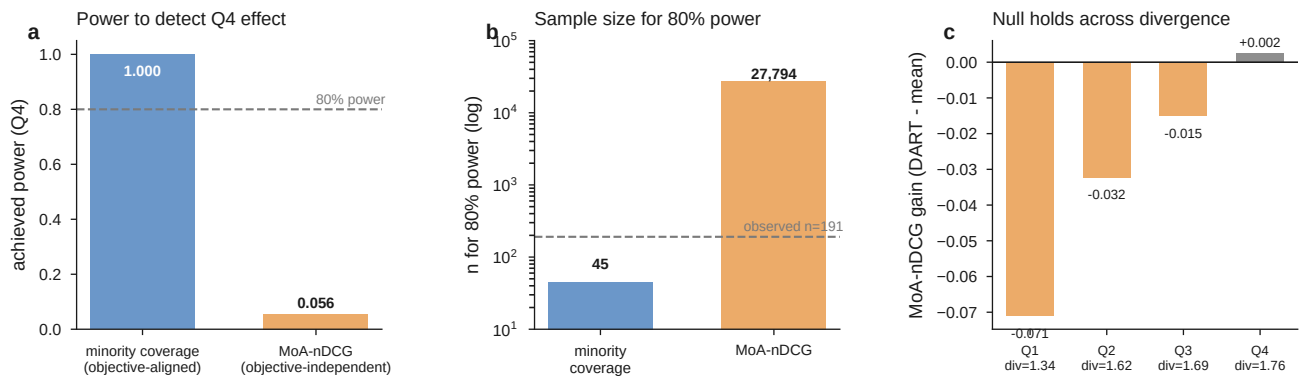
Supplementary Information: Objective-aligned evaluation inflates the distributional gain in single-cell drug retrieval

[[Author list, as in the main manuscript]]

This Supplementary Information contains six Extended Data figures, three Supplementary Notes, and four Supplementary Tables.

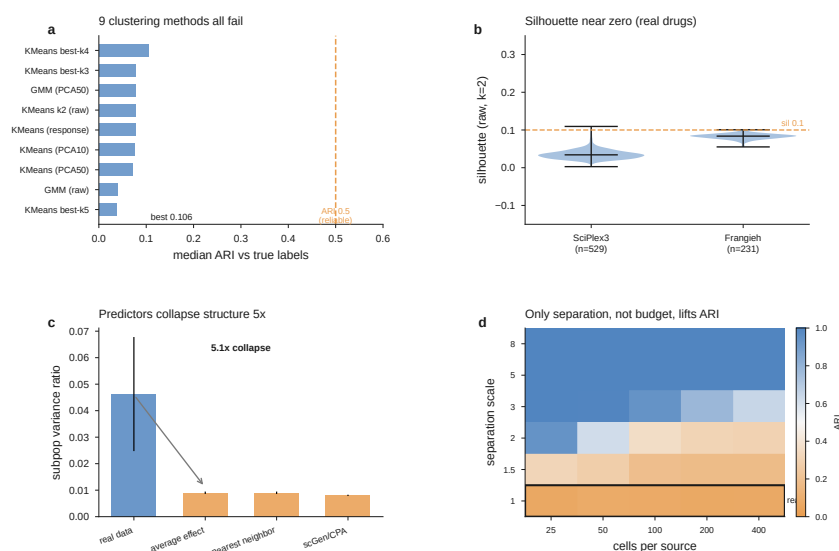


Extended Data Fig. 1: **Extended Data Fig. 1. Reproducibility basis.** **a**, Dataset scale across the four benchmarks (cells, perturbations). **b**, DART metric-family definitions. **c**, Spearman correlation among all retrieval metrics, computed over the 54,180 query–candidate scores generated by 1,260 retrieval queries (43 candidates each; the unit is the scored pair, not the query): mean cosine and CMap cosine are perfectly correlated ($\rho = 1.00$), while energy and coverage metrics form a separate, near-orthogonal block.

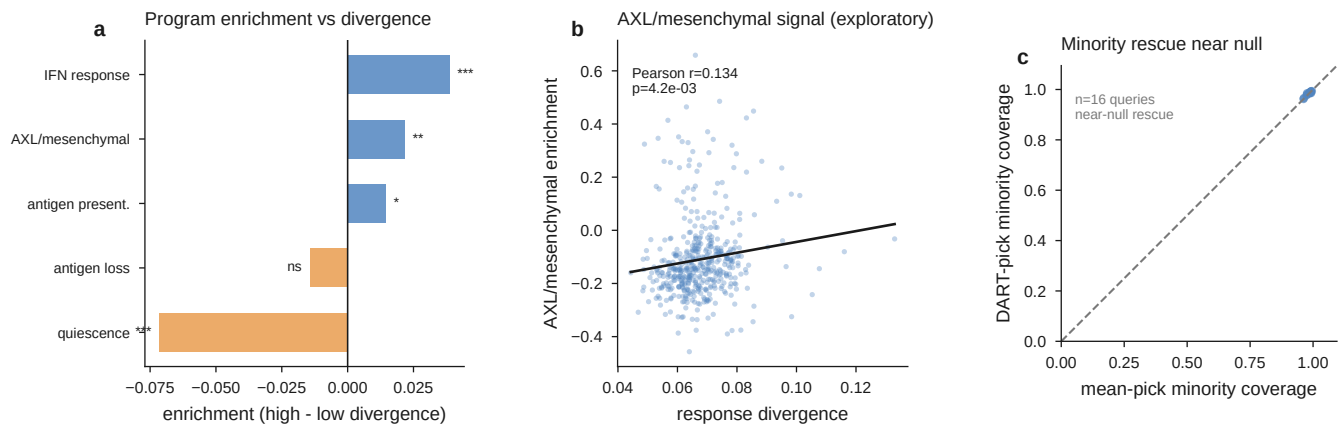


gain stays ≤ 0 across all divergence strata (not a power artifact)

Extended Data Fig. 2: **Extended Data Fig. 2. The Class B negative is not a power artifact.** **a**, Achieved power to detect the Q4 effect: the task-proximal minority-coverage metric is fully powered (1.00, $n=191$), the task-proximal MoA-nDCG is not (0.06, $n=143$). **b**, Sample size for 80% power: 45 versus 20,844 queries. **c**, MoA-nDCG gain by divergence stratum. It is significantly *negative* in Q1 (median -0.030 , Benjamini-Hochberg $q = 5.8 \times 10^{-5}$) and has a median of exactly zero in Q2–Q4, with no stratum showing a significant positive gain; the highest-divergence stratum's small positive *mean* ($+0.003$) is not significant ($q = 0.57$). MoA-nDCG is undefined for the leave-MoA-out and partial-library splits, where the mechanism class is removed from the library, so those 165 of 765 queries are excluded and $n=600$. A power deficit alone therefore does not explain the null: at the divergence level where the theory predicts the largest gain, the effect is absent, and where responses are least divergent it is adverse.

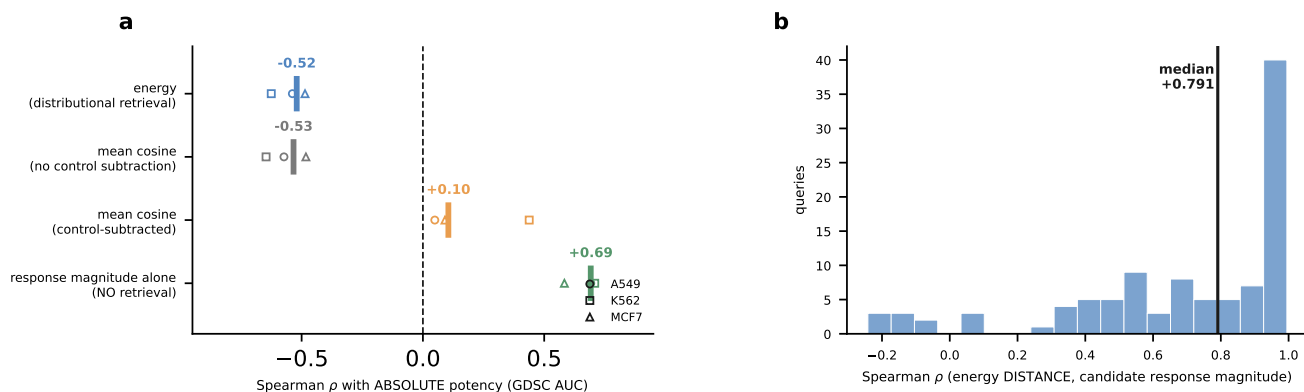


Extended Data Fig. 3: **Extended Data Fig. 3. Subpopulation identifiability in the constructed mixture, and why this panel alone could not settle it.** **a**, The original nine-column clustering sweep on a known bimodal mixture, reported here for completeness and superseded by main-text Fig. 5e. The best value across methods is a median adjusted Rand index of 0.106, an *upper bound* rather than a typical value: it is the best-case column taken over the minority of seeds in which that column wins, while the dense per-method columns lie at 0.037–0.078 and the global maximum over all methods and seeds is 0.149. The nine columns comprise only 6–7 *distinct* algorithms (the response-space $k=2$ column is bit-identical to its raw-expression counterpart because k -means is translation-invariant, and the three best- k columns are one method under three names), and *all are centroid- or Gaussian-based*. **This panel alone cannot support the conclusion once drawn from it.** Clustering failure is consistent both with the information being absent and with its being present but unreachable without labels, and those have opposite implications. Main-text Fig. 5e resolves it by adding the graph- and density-based methods missing here (Leiden, HDBSCAN) and, decisively, a supervised ceiling: the ceiling is 0.692 accuracy and the best unsupervised method reaches 0.674, so *within this constructed mixture* the limit is informational rather than algorithmic. **That qualifier turned out to be everything, and even the corrected reading was corrected again.** The 0.692 ceiling is largely an artefact of *pooling* several HDAC and several JAK compounds into two classes: split the same cells one drug against one drug and the ceiling rises to 0.879. Posed as the same drug-response question in a patient’s tumour, the ceiling is 0.923 while unsupervised clustering reaches only 0.777, a gap of +0.117 (main-text Fig. 6d): the information is present and the clustering step cannot reach it, so in real tissue the limit is *algorithmic*, not informational. An intermediate version of this correction compared the drug-response partition here with a *cell-type* partition in the tumour (malignant versus myeloid control cells, ceiling 0.964), which is a far easier question and licensed nothing; that number is withdrawn too. The general claim once drawn from this panel is withdrawn. **b**, Silhouette scores near zero across real drugs in both datasets (median 0.034; 0.4% of drugs exceed 0.1), consistent with the constructed-mixture reading. **c**, Structure diagnostics for predicted versus real candidate populations. Predicted populations *retain* baseline structure (subpopulation-variance ratio 0.142 versus 0.138 real); what they appear to lack is differential response (induced response cosine 0.19–0.37 versus 0.014 real). **That comparison does not survive scrutiny and is retained only for the structure-preservation point, which does not depend on it:** these candidate populations blend two cell lines, so an additive predictor emits a different delta per context, and for a latent model the response measured against real control cells is contaminated by the generator’s reconstruction bias. Measured within one context and against the model’s own baseline, an additive predictor returns exactly 1.000000 (main text; CORRECTIONS.md R28, R30). Predicted populations are synthesized by applying the predicted effect to the query context’s real control cells; synthesizing them instead as an isotropic Gaussian around a single mean would make every structure diagnostic a measurement of the synthesizer (isotropy index 0.998, variance ratio 0.009 by construction), see Methods. **d**, Identifiability phase diagram over separation $\times$ cell budget: only source separation lifts ARI, and the cell-budget axis is flat, so the failure is not a sample-size problem. The separation manipulation is fit once on the full source populations, independently of the sample being evaluated; fitting it on the sampled cells and then shifting those same cells amplifies whatever separation each small draw happens to contain and yields the indefensible result that ARI *falls* as the cell budget rises. See Methods.

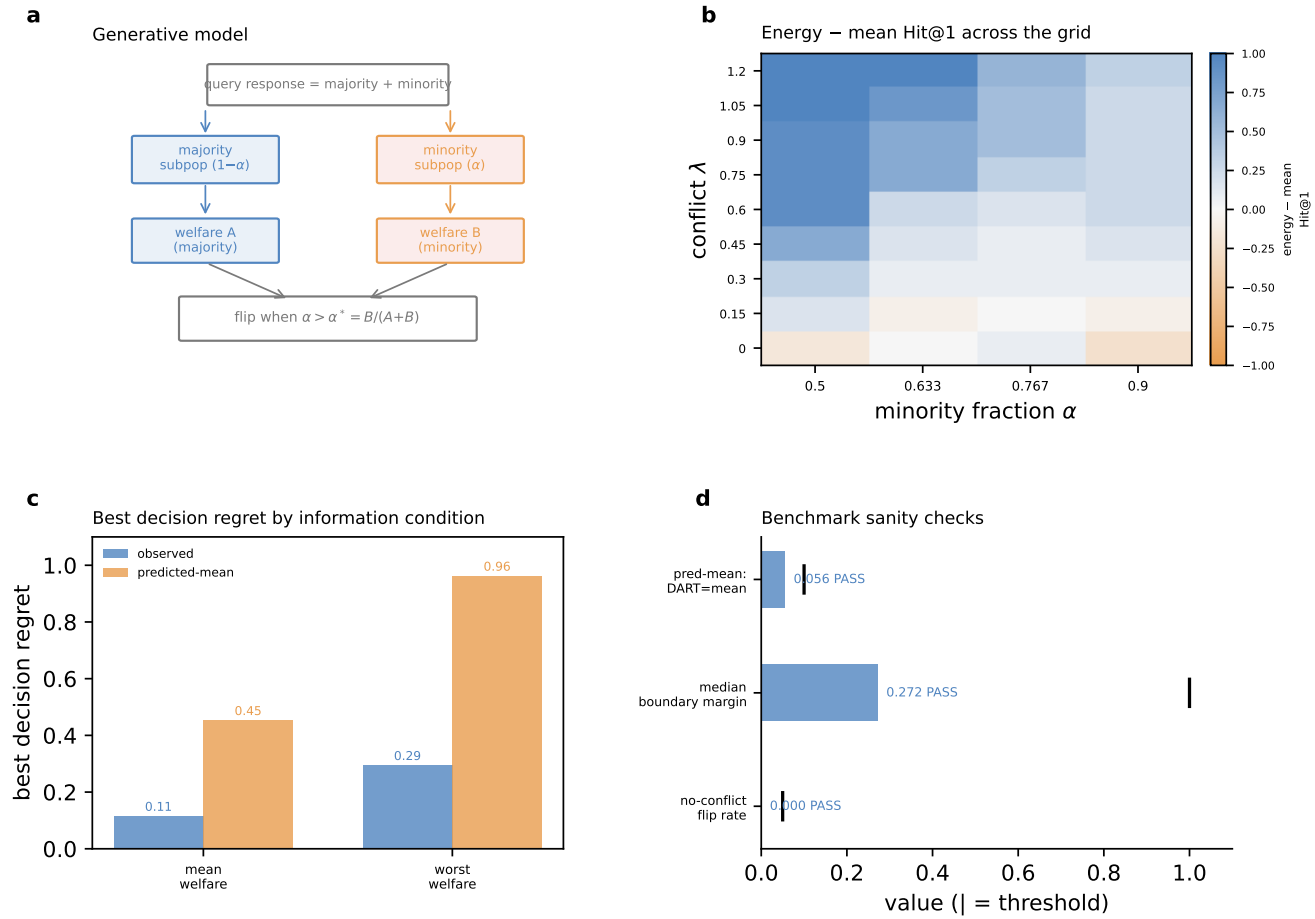


Exploratory (not claimed as robust); all signals carry qualifiers; no survival or timecourse validation

Extended Data Fig. 4: **Extended Data Fig. 4. Exploratory resistance-associated divergence.** **a**, Program enrichment in high- versus low-divergence minority states (IFN response $p = 1.9 \times 10^{-6}$, AXL/mesenchymal $p = 3 \times 10^{-3}$). **The largest-magnitude effect in this panel is neither of those, and we do not interpret it.** The quiescence program carries an enrichment of -0.072 , nearly twice the size of the next largest ($+0.039$ for IFN response), and it is highly significant on its face ($p = 7.7 \times 10^{-7}$, marked ***). Its *sign* is not stable across runs, so we report the bar rather than suppress it, and we draw no conclusion from it (Supplementary Note 3). Removing it because it is inconvenient would be the selection this paper exists to criticize; leaving it unlabelled would be worse. **b**, AXL/mesenchymal enrichment versus response divergence (Pearson $r = 0.134$). **c**, Minority-rescue contrast between DART and mean picks is near null. All signals are exploratory and hypothesis-generating, with no survival or timecourse validation.



Extended Data Fig. 5: Extended Data Fig. 5. The absolute-potency confounder audit, and why it is not a Class-C result. An earlier version of this work graded the retrieval rankings against *absolute* drug potency and concluded that retrieval similarity is anti-aligned with therapeutic utility. That conclusion is withdrawn, and this figure records both the analysis and the reason. **a**, Scored against absolute GDSC2 AUC, every retrieval ranking inverts (energy $\rho = -0.52$; a mean baseline without control subtraction, -0.53 ; the correctly specified incumbent, $+0.10$) while a query-independent scalar that ranks candidates by the magnitude of their own response and never looks at the query reaches $+0.69$. *The oracle is mismatched to the task.* A similarity retriever is asked which candidate resembles the query, not which kills hardest; handed a weak query it should correctly return other weak drugs, and this scoring counts that as failure. Because potency is largely driven by response magnitude, a magnitude scalar is near-guaranteed to win here. No therapeutic conclusion follows in either direction. The semantically matched oracle, drug–drug functional similarity, is in main-text Fig. 4h,i. **b**, What the audit does establish, and it is the reason the magnitude control is mandatory: the energy *distance* between a query and a candidate is largely a function of the candidate’s own response magnitude (median $\rho = +0.791$ across queries), because large-response populations sit far from everything. Ranking nearest-first therefore systematically returns the smallest-response candidates. Any distributional retrieval method claiming a functional validation must be shown to beat a scalar magnitude baseline.



Extended Data Fig. 6: **Extended Data Fig. 6. HIR-Bench: the panels demoted from the main text.** Main-text Fig. 6 keeps only the two HIR-Bench panels that carry argument, the analytic condition under which subpopulation structure can change a decision and the 2×2 that *measures* circularity. The four panels here were previously in the main text and should not have been: HIR-Bench’s phase boundary is designed into its generator, so recovering it verifies that the benchmark works rather than establishing anything about biology, and its transfer to real data fails (Supplementary Note 1). A reader could reasonably have read four main-text panels of a synthetic benchmark as biological evidence. They are retained because they document that the benchmark behaves as specified, which is a precondition for trusting the two panels that stayed. **a**, Generative model: a query response is a majority plus a minority subpopulation with welfares A and B. **b**, Energy-minus-mean Hit@1 across the (minority fraction, conflict) grid. **c**, Best decision regret by information condition. Collapsing candidates to their means inflates regret against both objectives and most severely against the worst-welfare one (0.29 observed versus 0.96 predicted-mean), which is the objective that depends on the minority subpopulation. **d**, Benchmark sanity checks: the no-conflict flip rate is 0.000 and the median boundary margin 0.27, both within threshold.

6 **Supplementary Note 1. HIR-Bench: sanity checks, and why the transfer test's 100% flag rate**
7 **is not evidence.**

8 The benchmark's internal checks hold on the full grid: the no-conflict flip rate is 0.0000, as it must be for
9 a generator whose flip boundary is $\alpha^* = B/(A + B)$, and the median boundary margin is 0.27.

10 Under the information-condition transfer test, the gate flags mean-collapsed predicted responses as
11 no-DART in 100% of cases. We decline to count that number as evidence, because it is a **tautology**: the
12 rule returns no-DART *unconditionally* whenever the information condition is predicted-mean. It cannot
13 return anything else, so the 100% restates the rule rather than testing it. The substantive claim, that
14 mean-collapsed candidate libraries offer no distributional advantage, rests on the predict-then-rank ex-
15 periments of the main text, not on this figure.

16 The projection of the HIR-Bench regime boundary onto real data **fails its own acceptance test, and**
17 **we report it as failing.** The label is heavily imbalanced (228 to 11), so a constant predictor scores 95.4%;
18 the regime projection scores 61.1%, which is 34 points *worse* than saying nothing. The boundary does
19 not transfer. Two structural limits bound the test in both directions and we work around neither: the
20 boundary is fitted on the method-performance layer, which is available on a 144-instance grid (72 grid
21 cells) because scoring nine retrieval methods against up to 100 candidates per instance does not scale to
22 the full grid; and the acceptance label derives from the same uncalibrated +0.01 threshold the main text
23 declines to trust.

24 **Supplementary Note 2. Two further tests of the two-gate criterion.**

25 **(a) Coverage performance versus query identifiability.** The two gates predict where a subpopulation-
26 coverage score should and should not carry signal. Stratifying real-data coverage by query identifiability
27 is consistent with them, but weakly and *not* monotonically. Coverage MoA-nDCG is higher for more
28 identifiable queries (Spearman $\rho = 0.17$, $p = 6.3 \times 10^{-5}$, $n = 543$ query-seed pairs), but the lift is
29 confined to the top tertile: tertile means are 0.285, 0.240 and 0.444 from least to most identifiable, so the
30 middle tertile sits *below* the least-identifiable one. Two caveats bound even that. The most-identifiable
31 tertile carries a median of 6 same-mechanism candidates against 2 in the other two tertiles, and a larger
32 positive pool mechanically raises nDCG, so the correlation is not cleanly separated from candidate-pool
33 size. And Gate 2 is almost never open *in these SciPlex3 queries*: even the most-identifiable tertile has a
34 median silhouette of 0.038. That scope matters. On patient glioblastoma the same gate is wide open
35 (main-text Fig. 6c), so the statement here is a fact about the cell-line data this note analyses and not about
36 biology.

37 **(b) The conditional advantage the criterion predicts is absent.** If identifiable structure is what lets
38 distributional scores help, the per-query mechanism-recovery advantage should turn positive where
39 structure is present. Stratifying 600 leave-drug-out queries by response divergence, it does rise, and the
40 ordering is real and leakage-free (Spearman $\rho = 0.14$, permutation $p = 2 \times 10^{-4}$). But it never clears
41 parity: the most-divergent stratum reaches only +0.02 in nDCG, with a bootstrap interval spanning
42 zero, and it reverses in one of three cell lines. Decisively, when queries are stratified by the actual Gate-2
43 variable, query identifiability (silhouette), the advantage is flat ($\rho \approx 0$). Response divergence measures
44 response spread, not resolvable subpopulation structure, and Gate 2 does not open *in these SciPlex3*
45 *queries* even for the most response-divergent ones. **This is not a statement about real tissue**, and an
46 earlier version of this note wrote it as though it were. On patient glioblastoma Gate 2 opens (main-text
47 Fig. 6c) and the distributional advantage still does not follow, which is why the two-gate criterion is
48 reported as necessary but not sufficient.

49 **Supplementary Note 3. Exploratory resistance-associated divergence.**

50 In the Frangieh melanoma Perturb-CITE-seq data, cells whose response direction diverges most between
51 immune contexts are enriched for resistance-associated programs: an AXL/mesenchymal program (cor-
52 relation +0.134, $p = 0.003$), an interferon-response program ($p = 1.9 \times 10^{-6}$), and antigen-presentation
53 genes ($p = 0.04$); see Extended Data Fig. 4.

54 This analysis is **exploratory and hypothesis-generating throughout** and is reported as a direction for
55 future work, not as a result. There is no longitudinal drug timecourse, no survival readout, and no post-
56 treatment clone measurement, so we make **no claim that a distributional score predicts resistance** or
57 identifies persister cells. A quiescence program's sign was not stable across runs and is not interpreted.
58 Consistent with the rest of the study, the distributional-versus-mean minority-rescue contrast is itself
59 near-null. The observation motivates future experiments with functional readouts, and nothing more.

60 **Supplementary Table 1. Datasets, preprocessing, and query construction.**

61 **(a) Dataset scale.** Four benchmarks span controlled proof, natural heterogeneity, an honest negative
 62 control, and a synthetic analytic stress test.

Dataset	Cells	Genes	Context axis	Perturbations
SciPlex3 (Srivatsan 2020)	276,325	2,000 HVG	A549, K562, MCF7	188 drugs (all 3 lines)
Frangieh (2021)	218,023	2,000 HVG	Control, IFN γ , Co-culture	239 CRISPR KOs
CD34+ (GSE306429)	33,984	2,000 HVG	discovered subpopulations	36 drugs
HIR-Bench (synthetic instances)	13,440	latent	two-subpopulation welfare model	tunable conflict
GDSC2 dose-response (Class C)	286 drugs $\times$ 969 lines		966 lines (SciPlex3 lines held out)	35 matched compounds

63 **(b) Preprocessing.** Processed tensors are built once and read directly by the retrieval code; the build
 64 scripts are kept for provenance.

Step	Setting
Source	SciPlex3: Srivatsan & Trapnell 2020; Frangieh: scPerturb Zenodo; CD34+: GSE306429 counts layer
Normalization	log-normalized expression; 2,000 highly variable genes selected per dataset
Controls	real DMSO controls (SciPlex3); condition controls (Frangieh); DMSO (CD34+)
Response delta	per-cell expression minus matched control centroid, used only where a metric requires it
Dose	10,000 nM (SciPlex3 controlled and cross-line experiments)

65 **(c) Query construction.** A query is a mixture of two response modes at mixing weight α ; the retrieval
 66 task is to rank the candidate that generated it above distractors.

Parameter	Controlled (SciPlex3)	Cross-line / natural
Response modes	HDAC (class a) vs JAK (class b)	cell-line pairs / immune-context pairs
Query size	400 cells	200 cells
Distractors	40	40
Mixing weight α	0.5, 0.6, 0.7, 0.8, 0.9	0.5–0.9
Seeds	20 (base 1000)	10 (cross-line), 20 (Frangieh)
Identifiability probe	min 30 cells, gate 0.9	min 30–40 cells, gate 0.9

67 Cell and gene counts are the processed-tensor totals. HIR-Bench’s 13,440 entries are synthetic benchmark *instances* (672 param-
 68 eter configurations $\times$ 20 seeds), not biological cells; they are the grid used for the predictability analysis (Fig. 6b).

69 **Supplementary Table 2. Field-level evaluation audit.**

70 The claim that objective-aligned evaluation is a field-wide default should be checkable rather than as-
 71 serted, so the audit is reproduced here rather than left with the code. **Inclusion criteria:** a method family
 72 qualifies if (i) it ranks or selects perturbations, or predicts perturbation responses that a ranking layer
 73 consumes, (ii) it is represented by at least one widely used published implementation, and (iii) its head-
 74 line evaluation metric is stated in the source. **Alignment** is judged *high* when the headline metric is a
 75 monotone transform of the scoring rule’s own objective evaluated on the same data, which is the condi-
 76 tion under which a gain is close to guaranteed. Families are the unit; we do not score individual papers,
 77 and this is an audit of *evaluation practice*, not of correctness.

Method family	Objective	Headline evaluation	Metric class	Alignment risk
CMap / L1000 sig- nature retrieval	rank by connectivity of mean differential signa- tures	connectivity score; cosine of mean signatures	A	high: the met- ric is the scoring rule restated
Single-cell signa- ture retrieval	rank by single-cell- derived mean or pseu- dobulk signature match	signature correlation; Hit@k; DEG overlap	A (B via DEG)	high: signature- space metric mir- rors signature- space scoring
Distributional re- trieval (this work)	rank by population- to-population distribu- tional distance	energy/MMD welfare- regret proxy (A); MoA- nDCG, minority coverage (B); GDSC functional similarity (C)	A + B + C	high on the Class-A proxy; <i>self-raised and measured here</i>
Single-cell pertur- bation predictors	predict post- perturbation expression from control	reconstruction MSE / R^2 / PCC of predicted delta	A (B via DEG)	high: training loss and evalu- ation share the reconstruction target
Virtual-cell / per- turbation founda- tion models	general-purpose predic- tion of perturbed cell state	held-out reconstruction; embedding similarity	A + B	high: recon- struction and embedding met- rics align with the pretraining objective
Graph / target- based inverse design	rank by graph proximity or target overlap	graph proximity; target- family overlap	A	high: the scoring graph and the evaluation graph coincide
Target/drug graph methods (PDGrapher-style)	predict interventions moving a cell state to- ward a target	closed-loop target rank- ing recall/nDCG	A (B via target family)	high: the closed- loop metric is aligned with the graph objective

78 The table audits **six published method families**, with the present work listed as a seventh row for contrast and excluded from
 79 every count below. Each of the six published families has a headline metric in Class A, and none reports a Class-C readout as
 80 its primary evidence. This is the pattern the present study raises against itself: we report all three classes on one task set, and
 81 the Class-A gain does not survive the other two at anything near its apparent size. The unit of the audit is the *family*, not the
 82 paper, and it is an audit of evaluation practice rather than of correctness; we do not score individual papers. Full citations with
 83 DOIs accompany the code.

84 **Supplementary Table 3. Cell-compartment validation in the natural-heterogeneity bench-**
85 **mark.**

86 The natural-heterogeneity result (main-text Fig. 6c–e) depends entirely on the cell-type labels being what
87 they claim to be, so the labelling is validated rather than trusted. Each cell receives a z-scored marker
88 score per compartment and is called only if it scores highest for that compartment, clears an absolute
89 expression floor on that compartment’s key markers, and beats the runner-up by a margin; everything
90 else is labelled unassigned and *dropped* (39.9% of cells). The table below reports, for each called com-
91 partment, the mean log-normalized expression of every compartment’s key marker set. A compartment
92 passes only if its own markers are the most strongly expressed. The loader raises rather than writing
93 labels if any compartment fails.

Called compartment	Cells	Mean expression of key markers for			Passes
		malignant	myeloid	oligodendrocyte	
Malignant glioma	41,314	1.024	0.073	0.117	yes
Tumour-assoc. myeloid	31,678	0.121	1.534	0.169	yes
Oligodendrocyte	23,233	0.409	0.114	1.784	yes

94 **What this table exists to prevent.** A first version of the assignment took a bare argmax over five marker sets, which forces
95 every cell into some compartment, including cells expressing none of the markers. It produced two compartments that fail
96 exactly this check: cells labelled T cells (22% of the dataset) had a mean CD3D/CD3E/CD2 expression of 0.080, *lower* than
97 their malignant-marker score of 0.141, and cells labelled endothelial expressed myeloid markers (1.074) more strongly than
98 endothelial ones (0.951). They were neither. Building a natural-heterogeneity result on a fabricated T-cell compartment would
99 have been worse than reporting no natural dataset at all, and only the three compartments above are used.

100 **Supplementary Table 4. The Class-C functional oracle, under both constructions.**

101 Cell lines differ greatly in general drug sensitivity, and that shared main effect inflates every drug–drug
 102 correlation: left uncorrected, 85% of drug pairs appear positively similar (median +0.178). The oracle
 103 used in the main text therefore subtracts each cell line’s mean AUC, estimated across all 286 GDSC2 com-
 104 pounds, before correlating; the corrected oracle is symmetric about zero (52% positive, median +0.015).
 105 **That choice decides the result, and the table below is why we state the dependence in the main text**
 106 **rather than reporting only the ordering we prefer.** Values are median per-query Spearman ρ against
 107 the oracle, over 103 leave-one-drug-out queries in three cell lines.

Ranking rule	Corrected oracle (cell-line mean removed)	Uncorrected oracle (raw AUC profiles)
Energy distance (the distributional score)	+0.276	+0.097
Mean cosine, control-subtracted (the incumbent)	+0.083	+0.138
Mean cosine, no control subtraction	+0.241	+0.092
Magnitude matching (query-dependent scalar)	+0.232	+0.217
Magnitude only (query-independent)	−0.330	+0.135
Potency matching (diagnostic, not a recommender)	+0.399	+0.290
Energy, partialled on magnitude matching	+0.097	n/a

108 **What flips and what does not.** Under the corrected oracle the distributional score leads its incumbent threefold (+0.276 against
 109 +0.083), and that is the one criterion in this study, invisible to every scorer, on which it wins. Under the uncorrected oracle it
 110 does not win at all: energy falls to +0.097, *below* the incumbent’s +0.138 and well below the magnitude scalar’s +0.217. The
 111 affirmative Class-C finding is therefore conditional on the centering. We think the centering is right, we specified it on the
 112 reasoning above before inspecting any retrieval result, and leaving a shared cell-line main effect inside an oracle would be an
 113 error in its own right. But a paper whose thesis is that the choice of criterion decides the winner cannot quietly exempt its own
 114 criterion, and an earlier draft of this work asserted that both oracles “give the same verdict”, which the table shows is false.
 115 **What survives both constructions is the negative result:** under neither oracle does the distributional score significantly beat
 116 a scalar that compares no distributions at all. Note also that energy’s uncorrected value (+0.097) and its magnitude-partialled
 117 value under the corrected oracle (+0.097) coincide to three decimals by arithmetic accident; they are different quantities (0.0967
 118 and 0.0969 before rounding).